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Thermodynamic study of the Al-Sc-Y system

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ABSTRACT

The phase equilibria of the Al-Sc-Y system at 873 K have been determined by scanning electron microscopy/energy dispersive X-ray spectroscopy and X-ray diffraction using individual alloys. Phase transition temperatures for some of the annealed samples have been measured by differential scanning calorimetry. Before establishing the thermodynamic description of the ternary system, the phase diagram of the Al-Y system was re-optimized, showing an improvement in the calculated enthalpy of formation of the compounds compared with previous assessments. Based on the experimental results of this work and the re-assessed Al-Y binary phase diagram, the thermodynamic description of the Al-Sc-Y ternary system was developed combining the first-principles calculation and the CALPHAD method. With the obtained thermodynamic parameters, the isothermal section at 873 K, the vertical sections at 9 at.% Sc and 16 at.% Y, and the liquidus projection are all calculated, which account for most of the experimental information satisfactorily.

1. Introduction

Since the addition of scandium was found to improve the mechanic properties of the aluminum alloys substantially at elevated temperatures, the pursuit of appropriate candidates of Sc has never been stopped in order to further improve the performance of the alloys while lowering the production cost. With the joint addition of Sc and Zr, the mechanic properties of the aluminum alloys are even superior than the Zr-free ones [1–6], which is attributed to the formation of the coherent $L1_2$_Al_3(Sc, Zr) precipitates in the matrix [7]. Yet, the ideal candidates should not only have a large solubility in the Al_3Sc phase, but also mildly increase the lattice parameter of the Al_3Sc phase [8]. In fact, the rare-earth elements like Er, Gd, Y etc. are all good choices for the substitution of Sc [9–15]. Karnesky et al. [15] investigated the influence of substituting rare-earth elements for Sc in the Al-0.08 at.% Sc alloy. As is indicated by the age hardening curves of the Al-0.06Sc-0.02 RE (at.%, RE = Dy, Er, Gd, Sm, Y or Yb) at 573 K, the peak hardness did not change much compared with that of the Al-0.08 at.% Sc alloy, but definitely higher than the Al-0.06 at.% Sc alloy.

In this work, the rare-earth element Y, which is adjacent to Sc in the IIIB subgroup of the periodic table of elements, is chosen as our target element, and the Al-Sc-Y system as our target system. The phase equilibrium, phase transition and phase diagram of this ternary system have been studied systematically via key experiments and a reasonable thermodynamic description has been established. It is hoped that this study would provide experimental and theoretical basis for the design of novel heat resistant aluminum alloys.

2. Review of the binary and ternary systems

2.1. Al-Y binary system

The phase diagram of the Al-Y binary system has been experimentally investigated by many authors, among whom Snyder [16] and Lundin and Klodt [17] made the phase relations basically clear. They reached an agreement on the existence of the six intermetallic compounds, namely αAl_3Y, βAl_3Y, Al_2Y, AlY, Al_2Y_3 and AlY_2, and the types of the invariant reactions, though there is a big difference in the temperatures of some invariant reactions. Later, Drits et al. [18], Yamshchikov et al. [19], and Kononenko and Golubev [20] reported this binary diagram of the Al-rich region, which has a good agreement with the experimental results reported by Snyder [16]. However, there are still disputes about the existence of the Al_3Y_5 and AlY_3 phases [21–23]. In order to establish a more accurate and reliable phase diagram, Liu et al. [24] carried out a systematic experimental investigation on the phase equilibria of this system over the entire composition via differential thermal analysis (DTA), X-ray diffraction (XRD) and scanning electron microscopy/energy dispersive X-ray spectroscopy (SEM/EDX).

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The six stable phases as mentioned above were confirmed to be existed, while the Al_3Y_5 and AlY_3 phases were not observed. The entire phase diagram measured by Liu et al. [24] agrees very well with that reported by Snyder [16] and is thus mainly considered in the re-optimization of this work.

Besides the phase equilibria data, comprehensive thermodynamic properties of this binary system have been reported. Esin et al. [25] measured the partial mixing enthalpy of the liquid phase over 0–60 at.% Y at 1873 K with the high temperature calorimetry, and the enthalpy of mixing of the liquid phase was then calculated. Using the same method, Ryss et al. [26] determined these properties of the liquid phase over the entire composition at the same temperature. Lee and Sommer [27] measured the partial mixing enthalpy of Y in the melt containing 0.04–1.19 at.% Y at 1213 K with the high temperature solution calorimetry. Subsequently, the activities of Al and Y at 1873 K were calculated by Petrusheveskii and Ryss [28] based on the experimental mixing enthalpy reported by Ryss et al. [26]. The enthalpies of formation for some of the Al-Y binary compounds have been determined by Snyder [16], Jung et al. [29], Meschel and Kleppa [30], Timofeev et al. [31] and Borzone et al. [32]. Later, the enthalpies of formation of five Al-Y compounds, namely αAl_3Y, Al_2Y, AlY, Al_2Y_3 and AlY_2, were calculated by Tao et al. [33] employing the first-principles calculation method. The calculation results and the experimental data mentioned above are consistent among each other, except the formation enthalpies of Al_2Y and AlY phases measured by Snyder [16] that are too negative compared with others [28,29,31,32].

With the update of the experimental data, the Al-Y binary phase diagram was thermodynamically assessed by Ran et al. [34], Gröbner et al. [35], Lukas et al. [36], Shakhshir et al. [37], Liu et al. [38] and Kang et al. [39] successively. The standard enthalpies of formation of the compounds optimized by the Ran et al. [34] and Lukas et al. [36] are too negative. Besides, in the assessment of Lukas et al. [36], the Al_3Y_5 phase was taken into account, which has already been confirmed as a non-equilibrium phase. Though Gröbner et al. [35] made an improvement in the assessment of the formation enthalpies of the compounds, too many thermodynamic parameters were used for the liquid phase as is the same case in the work of Ran et al. [34] and Lukas et al. [36]. In the assessment of Shakhshir et al. [37], the optimized integral and partial enthalpies of mixing of the liquid phase have a positive deviation from the experimental data. Kang et al. [39] employed the modified quasichemical solution model to depict the liquid phase. Besides, the AlY phase was treated as the congruent melting compound as was done by Gröbner et al. [35]. As a matter of fact, it is a product of a peritectic reaction confirmed by experiments. On the whole, the phase diagram optimized by Liu et al. [38] agrees very well with the experimental data. However, the assessed standard formation enthalpies for all the compounds excluding Al_3Y have slightly large negative deviation from the experimental data. In order to establish an accurate thermodynamic description of the Al-Sc-Y ternary system, it is necessary to re-assess the Al-Y binary system.

2.2. Sc-Y binary system

Beaudry and Daane [40] investigated the Sc-Y binary phase diagram via thermal analysis and XRD. Since the temperatures for the melting points and the β↔α transition of pure elements disagree with the widely accepted values, Gschneidner and Calderwood [41] made a modification later. There are three phases in this system, including liquid, α(Sc, Y) and β(Sc, Y). The liquidus and solidus reach a common minimum at 1638 K, 50 at.% Y. The phase transition lines of β↔α reach a common minimum at 1453 K, 43 at.% Y. For the thermodynamic properties of this system, only the theoretical calculation data are available at present. With the Miedama model, Niessen et al. [42] predicted that the solution enthalpy of Sc in the dilute Y solution was 3 kJ/mol, and that of Y in the dilute Sc solution 4 kJ/mol. Wang et al. [43] have carried out a thermodynamic assessment on the Sc-Y binary phase diagram and their results are adopted in this work.

Table 1
Crystallographic data of solid phases in the Al-Sc-Y ternary system.

Phase	Lattice parameters			Prototype, Pearson symbol, space group (No.)	Ref.
	a (nm)	b (nm)	c (nm)		
Al	0.4050	0.4050	0.4050	Cu, cF4, Fm–3 m (225)	[45]
αSc	0.33088	0.33088	0.52680	Mg, hP2, $P6_3/mmc$ (194)	[46]
βSc	0.373	0.373	0.373	W, cI2, Im–3 m (229)	[46]
αY	0.36474	0.36474	0.57306	Mg, hP2, $P6_3/mmc$ (194)	[46]
βY	0.407	0.407	0.407	W, cI2, Im–3 m (229)	[46]
Al_3Sc	0.410	0.410	0.410	$AuCu_3$, cP4, Pm–3 m (221)	[45]
Al_2Sc	0.7600	0.7600	0.7600	Cu_2Mg, cF24, Fd–3 m (227)	[45]
AlSc	0.3450	0.3450	0.3450	ClCs, cP2, Pm–3 m (221)	[45]
$AlSc_2$	0.4888	0.4888	0.6166	$InNi_2$, hP6, $P6_3/mmc$ (194)	[45]
αAl_3Y	0.6276	0.6276	0.4582	Ni_3Sn, hP8, $P6_3/mmc$ (194)	[45]
βAl_3Y	0.6204	0.6204	2.1184	$BaPb_3$, hR12, R–3 m (166)	[45]
Al_2Y	0.78611	0.78611	0.78611	Cu_2Mg, cF24, Fd–3 m (227)	[45]
AlY	0.3884	1.1522	0.4385	BCr, oC8, Cmcm (63)	[45]
Al_2Y_3	0.8239	0.8239	0.7648	Al_2Zr_3, tP20, $P4_2/mnm$ (136)	[45]
AlY_2	0.659	0.512	0.930	Co_2Si, oP12, Pnma (62)	[45]
$Al_3(Sc, Y)_5$	–	–	–	–	this work
τ_1*	–	–	–	–	this work

* τ_1 represents the ternary compound $Al_{75}Sc_{17}Y_8$.

2.3. Al-Sc-Y ternary system

The phase relations of the Al-Sc-Y ternary system at 773 K over 75–100 at.% Al have been studied by Zalutskaya et al. [44]. According to their work, there may be a ternary compound Al_6ScY containing 5–13.5 at.% Sc, where the amount of Al keeps constant and the elements Sc and Y could substitute for each other. The maximum solubility of Sc in the αAl_3Y phase was reported as 4 at.%, while that of Y in the Al_3Sc phase 10 at.%. The crystallographic data of all the solid phases in this ternary system are listed in Table 1 [45,46], where τ_1 represents the ternary compound $Al_{75}Sc_{17}Y_8$ found in this work herein and after.

3. Experimental procedure

The pure metals Al, Sc and Y with a purity of 99.99 wt.% were used as starting materials, of which the specifications are listed in Table 2. Twenty samples, 4 g for 1–8# and 4.5 g for 9–20#, were arc-melted in the water-cooled copper crucible under the protection of argon of 0.06 MPa. With a Ti ball as the oxygen getter, they were melted four times and flipped after each round to ensure homogeneity. The nominal compositions are listed in Table 3, along with the compositions analyzed by EDX. Making a comparison between the nominal and analyzed atomic percent of the constituent elements, it can be seen that the difference between them is less than 2.5 at.% for most samples, while the maximum difference reaches 3 at.% for the samples 4#, 14# and 17#.

The as-cast samples were cut into two pieces. One of them was examined by FEI Quanta-200 SEM/EDX to have the primary crystallization phase identified. The other was sealed in the vacuumed and argon refilled quartz tube, and was subjected to 1440 h of annealing at 873 K, followed by a water quench. Then, all the annealed samples were examined by FEI Quanta-200 SEM/EDX and D/Max2500DC XRD to have the equilibrium constituent phases determined. The SEM/EDX

Table 2
Specifications of starting materials used in the experiments.

Chemical name	Source	Purity (wt.%)	Purification method	Analysis method
Aluminum	China New Metal Materials Technology Co., Ltd.	99.99	None	Certified purity
Scandium	China New Metal Materials Technology Co., Ltd.	99.99	None	Certified purity
Yittrium	China New Metal Materials Technology Co., Ltd.	99.99	None	Certified purity
Indium[a]	Netzsch-Gerätebau GmbH	99.999	None	Certified purity
Bismuth[a]	Netzsch-Gerätebau GmbH	99.999	None	Certified purity
Zinc[a]	Netzsch-Gerätebau GmbH	99.999	None	Certified purity
Aluminum[a]	Netzsch-Gerätebau GmbH	99.999	None	Certified purity
Aurum[a]	Netzsch-Gerätebau GmbH	99.999	None	Certified purity

[a] These materials were used for DSC calibration.

Table 3
Nominal and measured compositions of individual alloys.[a]

Alloy No.	Nominal composition			Composition measured by EDX at T = 298 K, P < 1 × 10^{-3} Pa		
	x_{Al} (atomic fraction)	x_{Sc} (atomic fraction)	x_Y (atomic fraction)	x_{Al} (atomic fraction)	x_{Sc} (atomic fraction)	x_Y (atomic fraction)
1#	0.800	0.155	0.045	0.811	0.135	0.054
2#	0.800	0.100	0.100	0.793	0.092	0.115
3#	0.800	0.045	0.155	0.804	0.036	0.160
4#	0.730	0.135	0.135	0.750	0.105	0.145
5#	0.691	0.253	0.056	0.704	0.234	0.062
6#	0.700	0.200	0.100	0.695	0.205	0.100
7#	0.698	0.137	0.165	0.703	0.128	0.169
8#	0.700	0.080	0.220	0.705	0.067	0.228
9#	0.635	0.183	0.182	0.646	0.161	0.193
10#	0.590	0.080	0.330	0.589	0.070	0.341
11#	0.545	0.300	0.155	0.559	0.277	0.164
12#	0.528	0.165	0.307	0.525	0.149	0.326
13#	0.515	0.155	0.330	0.524	0.131	0.345
14#	0.400	0.270	0.330	0.407	0.238	0.355
15#	0.410	0.224	0.366	0.417	0.199	0.384
16#	0.400	0.100	0.500	0.394	0.084	0.522
17#	0.273	0.300	0.427	0.281	0.260	0.459
18#	0.340	0.100	0.560	0.336	0.082	0.582
19#	0.310	0.100	0.590	0.303	0.085	0.612
20#	0.240	0.100	0.660	0.249	0.081	0.670

[a] Standard uncertainties u are u(T) = 1 K, u(P) = 0.0001 Pa, u(x_{Al}) = 0.002, u(x_{Sc}) = 0.012, u(x_Y) = 0.008.

tests were carried out under vacuum (< 1 × 10^{-3} Pa) at 298 K, with an accelerating voltage of 20 kV and a beam spot of 5.0 or 6.0 Å. Using the Cu Kα radiation, the XRD tests were performed in air atmosphere at 298 K and 101 kPa. The scanning ranged from 10° to 90° at a rate of 8°/min, with the voltage and the current set as 40 kV and 250 mA, respectively. Fifteen annealed samples, namely 1–4#, 6#, 8–9#, 11#, 13–14# and 16–20#, were also tested by the differential scanning calorimetry (DSC) on the NETZSCH STA 449C equipment. With the samples placed in the Al_2O_3 crucible covered by a lid also made of Al_2O_3, the heating started from room temperature to 1573 K for the samples 1–3# and 18–20# or to 1673 K for the others at a rate of 10 K/min, followed by a cooling to 473 K for all of them at the same rate. During the whole testing process, argon flowed through the chamber continuously at a rate of 25 ml/min to protect the samples from oxidation. Before performing the DSC tests, the equipment was calibrated using a series of standard materials of known melting points (denoted as T) including In, Bi, Zn, Al and Au all with a purity of 99.999 wt.%, which were supplied by Netzsch-Gerätebau GmbH. They were heated from room temperature at a rate of 30 K/min to about 100 K lower than the melting point (denoted as T_1), holding for 2 min, and then were heated to the temperature that was about 100 K higher than the melting point (denoted as T_2) at a rate of 10 K/min, holing for 2 min, followed by a cooling to T_1 at a rate of 10 K/min, holding for 2 min, and then a heating to T_2 at a rate of 10 K/min. At last, the samples were cooled naturally in the chamber. For the sample In, T_1 and T_2 were about 50 K lower and higher than the melting point, respectively. The onset points on the curves of the second heating were used as the measured melting points. The difference between the measured and theoretical melting points, denoted as ΔT, could be calculated. By fitting the data of (ΔT, T), a correction curve of ΔT versus T was then obtained. The temperatures

Table 4
Equilibrium phase constitution and compositions of constituent phases measured by EDX at T = 298 K, P < 1 × 10^{-3} Pa.[a]

Alloy No.	Phase	x_{Al} (atomic fraction)	x_{Sc} (atomic fraction)	x_Y (atomic fraction)
1#	fcc_Al	0.994	0.005	0.001
	τ_1	0.762	0.174	0.064
	αAl_3Y	0.750	0.037	0.213
2#	fcc_Al	0.978	0.015	0.007
	τ_1	0.760	0.167	0.073
	αAl_3Y	0.739	0.047	0.214
3#	fcc_Al	0.993	0.004	0.003
	αAl_3Y	0.745	0.047	0.208
4#	τ_1	0.759	0.159	0.082
	αAl_3Y	0.746	0.053	0.201
5#	Al_3Sc	0.753	0.211	0.036
	Al_2Sc	0.670	0.268	0.062
	Al_2Y	0.673	0.195	0.132
6#	Al_3Sc	0.761	0.201	0.038
	τ_1	0.754	0.167	0.079
	Al_2Y	0.674	0.199	0.127
7#	τ_1	0.744	0.171	0.085
	Al_2Y	0.678	0.078	0.244
8#	τ_1	0.762	0.143	0.095
	αAl_3Y	0.746	0.057	0.197
	Al_2Y	0.672	0.059	0.269
9#	Al_2Y	0.666	0.148	0.186
	AlSc	0.505	0.405	0.090
10#	Al_2Y	0.650	0.033	0.317
	AlY	0.495	0.110	0.395
11#	Al_2Y	0.650	0.151	0.199
	AlSc	0.493	0.384	0.123
12#	Al_2Y	0.655	0.091	0.254
	AlY	0.495	0.160	0.345
13#	Al_2Y	0.654	0.072	0.274
	AlY	0.496	0.141	0.363
14#	AlY	0.491	0.205	0.304
	Al_3Y_5	0.378	0.247	0.375
15#	AlY	0.490	0.164	0.346
	Al_3Y_5	0.373	0.213	0.414
16#	Al_2Y_3	0.397	0.140	0.463
	Al_3Y_5	0.386	0.054	0.560
17#	AlY_2	0.333	0.237	0.430
	hcp_(Sc,Y)	0.052	0.344	0.604
18#	Al_3Y_5	0.365	0.105	0.530
	AlY_2	0.322	0.076	0.602
19#	AlY_2	0.322	0.072	0.606
	hcp_Y	0.054	0.024	0.922
20#	AlY_2	0.323	0.085	0.592
	hcp_Y	0.028	0.031	0.941

[a] Standard uncertainties u are u(T) = 1 K, u(P) = 0.0001 Pa, u(x_{Al}) = 0.002, u(x_{Sc}) = 0.012, u(x_Y) = 0.008.

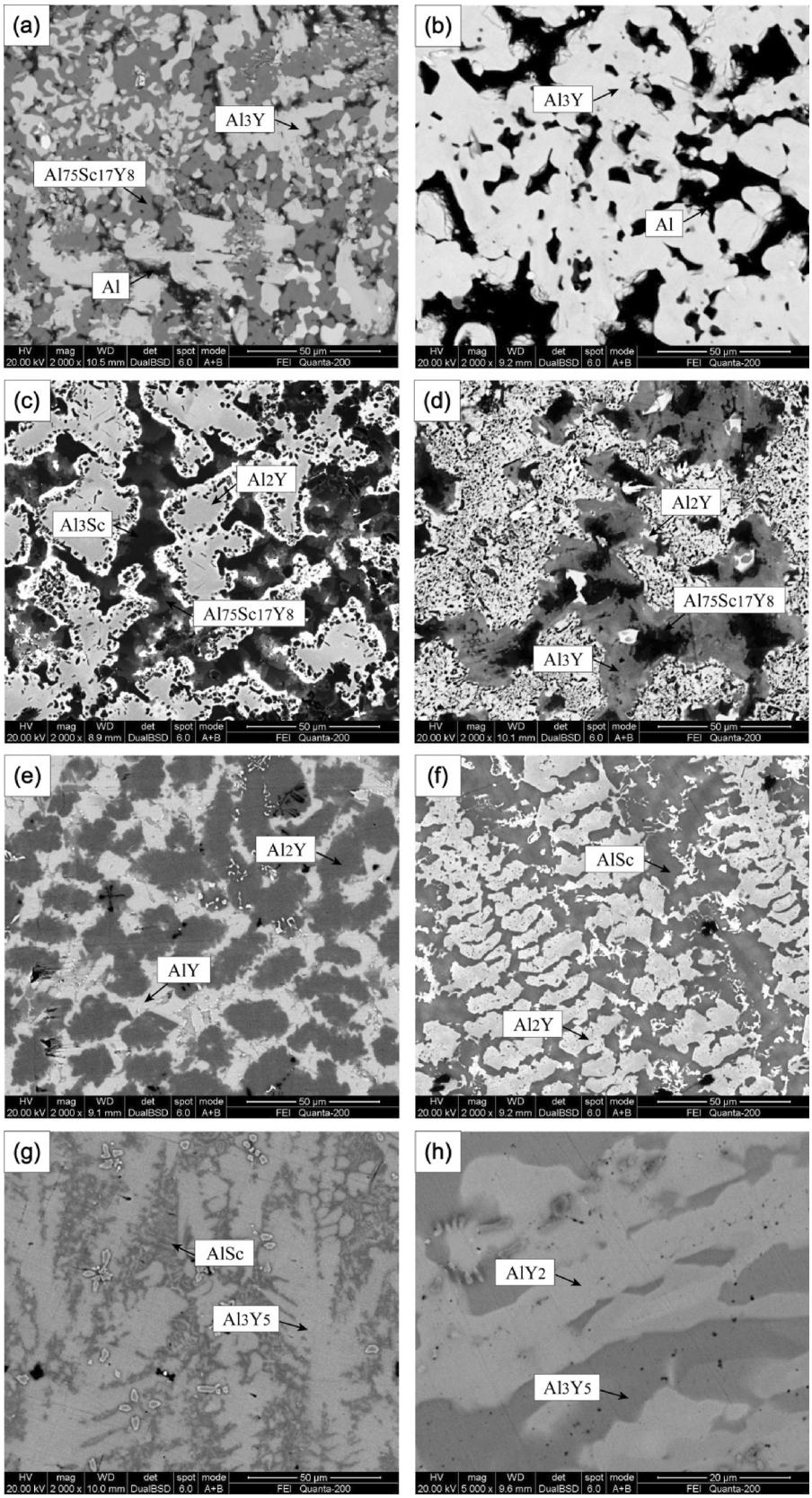

Fig. 1. Backscattered electron images of the annealed samples: (a) 2#; (b) 3#; (c) 6#; (d) 8#; (e) 10#; (f) 11#; (g) 14#; (h) 18#.

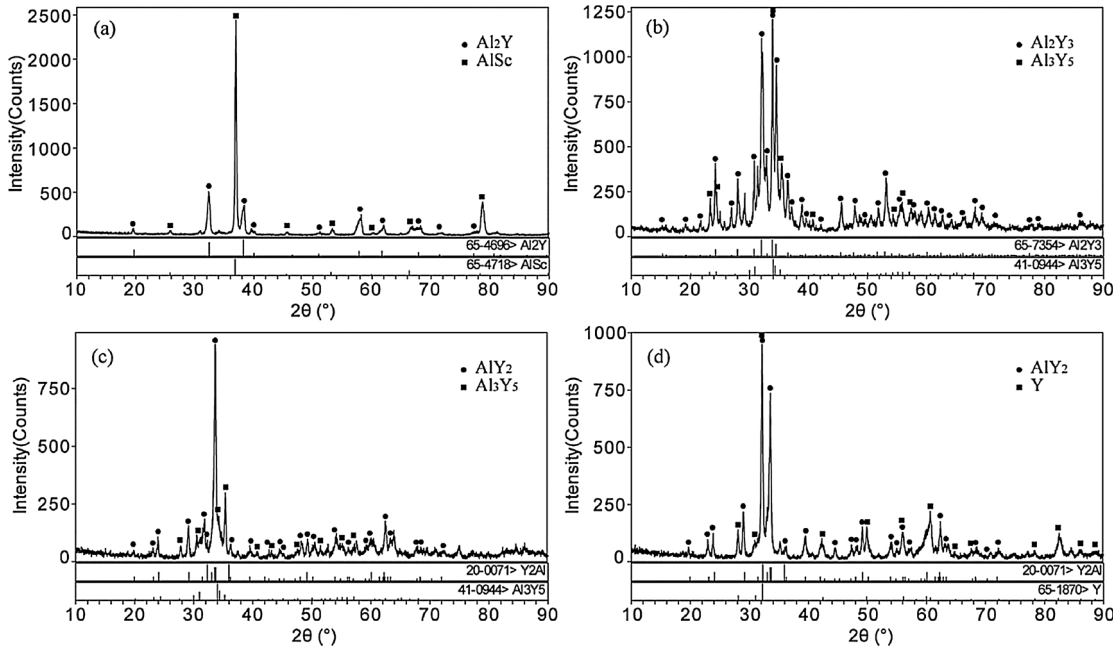

Fig. 2. XRD pattern of the annealed samples: (a) 11#; (b) 16#; (c) 18#; (d) 20#.

measured by DSC in this work are all calibrated.

4. Calculation methods

The Gibbs energies for the pure elements are taken from the SGTE unary database [47]. For all the solution phases in the constituent binary systems and the ternary system, including liquid, fcc_A1, bcc_A2 and hcp_A3, the substitutional solution model was employed to depict their Gibbs energy. Since there is a certain solubility for Sc in the Al-Y binary compounds, i.e. Al_3Y, AlY, Al_2Y_3 and AlY_2 and for Y in the Al-Sc binary compounds, i.e. Al_3Sc, Al_2Sc and AlSc, they are modeled as $Al_x(Sc, Y)_y$. The Gibbs energy per formula unit could be expressed as follows.

$$G_{Al_x(Sc,Y)_y} = y''_{Sc}G_{Al:Sc} + y''_{Y}G_{Al:Y} + \mathrm{yRT}(y''_{Sc}\ln y''_{Sc} + y''_{Y}\ln y''_{Y}) + y''_{Sc}y''_{Y}\left(\sum_{j=0,1\ldots} {}^{j}L_{Al:Sc,Y}(y''_{Sc} - y''_{Y})^{j}\right) \quad (1)$$

where y''_{Sc} and y''_{Y} represent the site fraction of Sc and Y in the second sublattice, respectively. $G_{Al:Sc}$ and $G_{Al:Y}$ could be formalized as Eqs. (2) and (3).

$$G_{Al:Sc} = x^0G_{Al}^{fcc} + y^0G_{Sc}^{hcp} + A + BT \quad (2)$$

$$G_{Al:Y} = x^0G_{Al}^{fcc} + y^0G_{Y}^{hcp} + C + DT \quad (3)$$

When describing an Al-Sc binary compound, $G_{Al:Sc}$ represents the Gibbs energy of Al_xSc_y per formula unit, and $G_{Al:Y}$ denotes that of a fictitious phase Al_xY_y with the structure of Al_xSc_y. The value of C is adopted from the first-principles calculation result and D needs to be optimized in this work. It is similar when modeling an Al-Y binary compound. The first principles calculation was carried out by the VASP (Vienna Ab initio Simulation Package) [48,49] software. The interaction between ions and electrons and the exchange-correlation energy were depicted by the all-electron projector augmented wave (PAW) function [50,51] and the generalized gradient approximation (GGA) [52,53], respectively. Al, Sc_sv and Y_sv are the potential functions used in this work. More details on the first-principles calculation could be found in our previous work [54].

The solubility of Y in the $AlSc_2$ phase is not considered for now since it is not determined in our experimental work. For the C15_Laves phase Al_2Y, it is described as $(Al, Y)_2(Al, Y)$ in the Al-Y binary system. Since the third element Sc could substitute Y, $(Al, Sc, Y)_2(Al, Sc, Y)$ is used when optimizing the Al-Sc-Y ternary system. The Gibbs energy per formula unit could be expressed as follows.

$$\begin{aligned} G_{(Al,Sc,Y)_2(Al,Sc,Y)} &= \sum_i\sum_j y'_i y''_j G_{i:j} + 2RT\sum_i y'_i \ln y'_i + \mathrm{RT}\sum_i y''_i \ln y''_i \\ &+ \sum_i\sum_{j>i} y'_i y'_j \sum_k y''_k\left(\sum_{n=0,1\ldots} {}^{n}L_{i,j:k}(y'_i - y'_j)^n\right) \\ &+ \sum_i\sum_{j>i} y''_i y''_j \sum_k y'_k\left(\sum_{n=0,1\ldots} {}^{n}L_{k:i,j}(y''_i - y''_j)^n\right) \\ &+ \sum_i\sum_{j>i}\sum_k\sum_{l>k} y'_i y'_j y''_k y''_l L_{i,j:k,l} \end{aligned} \quad (4)$$

where i, j, k and l represent components Al, Sc and Y. when $i = j$,

$$G_{i:j} = 3^0G_i + 15000 \quad (5)$$

when $j > i$,

$$G_{i:j} = 2^0G_i + {}^0G_j + m + nT \quad (6)$$

$$G_{j:i} = 3^0G_i + 3^0G_j + 30000 - G_{i:j} \quad (7)$$

where i denotes the element Al, and j denotes the elements Sc and Y. The parameter m in the expression of $G_{Al:Sc}$ was taken from the first-principles results in this work. Other m and n parameters need to be optimized.

Based on the experimental results of this work, there is a ternary compound $Al_{75}Sc_{17}Y_8$ in the Al-rich corner, which is modeled as $Al_xSc_yY_z$. The Gibbs energy per formula unit could be expressed as Eq. (8).

$$G_{Al_xSc_yY_z} = x^0G_{Al}^{fcc} + y^0G_{Sc}^{hcp} + z^0G_{Y}^{hcp} + p + qT \quad (8)$$

where p and q are the parameters to be optimized in the present work.

Table 5
Measured and calculated phase transformation temperatures and corresponding calculated invariant reactions or phase boundaries.[a]

Alloy No.	Measured temperature $T_{mea.}$ (K)	Calculated temperature $T_{cal.}$ (K)	Invariant reaction or phase boundary
1#	–	1543	Liquid + Al_2Y/Liquid + Al_2Sc + Al_2Y
	–	1540	Liquid + Al_2Sc + $Al_2Y \leftrightarrow \tau_1$
	1477	1532	Liquid + Al_2Y + τ_1/Liquid + τ_1
	1384	1468	Liquid + τ_1/Liquid + Al_3Sc + τ_1
	912	931	Liquid + $Al_3Sc \leftrightarrow$ fcc_A1 + τ_1
2#	1428	1474	Liquid + Al_2Y/Liquid + Al_2Y + τ_1
	1098	1131	Liquid + $Al_2Y \leftrightarrow \tau_1$ + βAl_3Y
	–	974	Liquid + τ_1 + $\beta Al_3Y \leftrightarrow \alpha Al_3Y$
	911	914	Liquid + $\tau_1 \leftrightarrow \alpha Al_3Y$ + fcc_A1
3#	1289	–	–
	–	1150	Liquid + Al_2Y/Liquid + Al_2Y + βAl_3Y
	1101	1131	Liquid + $Al_2Y \leftrightarrow \tau_1$ + βAl_3Y
	–	974	Liquid + τ_1 + $\beta Al_3Y \leftrightarrow \alpha Al_3Y$
	–	973	Liquid + αAl_3Y + βAl_3Y/Liquid + αAl_3Y
	–	914	Liquid + αAl_3Y/Liquid+ fcc_A1 + αAl_3Y
	913	913	Liquid+ fcc_A1 + αAl_3Y/fcc_A1 + αAl_3Y
4#	1448	1479	Liquid + Al_2Y/Liquid + Al_2Y + τ_1
	1287	–	–
	1093	1131	Liquid + $Al_2Y \leftrightarrow \tau_1$ + βAl_3Y
	–	974	Liquid + τ_1 + $\beta Al_3Y \leftrightarrow \alpha Al_3Y$
	911*	914	Liquid + $\tau_1 \leftrightarrow \alpha Al_3Y$ + fcc_A1
6#	1616*	1628	Liquid + Al_2Sc/Liquid + Al_2Sc + Al_2Y
	1511	1540	Liquid + Al_2Sc + $Al_2Y \leftrightarrow \tau_1$
	–	1522	Al_2Sc + $\tau_1 \leftrightarrow Al_3Sc$ + Al_2Y
	–	882	Al_2Y + Al_3Sc + Al_2Sc/Al_2Y + Al_3Sc
8#	1274	1147	Liquid + Al_2Y/Liquid + Al_2Y + βAl_3Y
	–	1131	Liquid + $Al_2Y \leftrightarrow \tau_1$ + βAl_3Y
	–	1025	τ_1 + Al_2Y + βAl_3Y/Al_2Y + βAl_3Y
	–	974	Al_2Y + βAl_3Y/Al_2Y + αAl_3Y + βAl_3Y
	–	967	Al_2Y + αAl_3Y + βAl_3Y/Al_2Y + αAl_3Y
9#	–	1579	Liquid + Al_2Y/Liquid + Al_2Y + AlY
	–	1566	Liquid $\leftrightarrow$ AlSc + Al_2Y + AlY
11#	1641*	1671	Liquid/Liquid + Al_2Y
	–	1628	Liquid + Al_2Y/Liquid + Al_2Y + Al_2Sc
	1539	1567	Liquid + Al_2Y + Al_2Sc/Al_2Y + Al_2Sc + AlSc
	–	1562	Al_2Y + Al_2Sc + AlSc/Al_2Y + AlSc
	1394	–	–
	1345	1352	Al_2Y + AlSc/Al_2Y + AlSc + AlY
13#	1608*	1630	Liquid/Liquid + AlY
	–	1629	Liquid + AlY/Liquid + AlY + Al_2Y
	1601*	1626	Liquid + AlY + Al_2Y/AlY + Al_2Y
	1280	–	–
	1258	–	–
14#	–	1533	Liquid/Liquid + AlY
	–	1470	Liquid + AlY/Liquid + AlY + Al_3Y_5
	1276	1433	Liquid + AlY + Al_3Y_5/AlY + Al_3Y_5
	1247	–	–
16#	–	1464	Liquid/Liquid+ AlY
	–	1462	Liquid + AlY/Liquid + AlY + Al_3Y_5
	1304	1432	Liquid + AlY + Al_3Y_5/AlY + Al_3Y_5 + Al_2Y_3
	1261	1359	AlY + Al_3Y_5 + Al_2Y_3/Al_3Y_5 + Al_2Y_3
	–	843	Al_2Y_3 + Al_3Y_5/Al_2Y_3 + Al_3Y_5 + AlY_2
	–	813	Al_2Y_3 + Al_3Y_5 + AlY_2/Al_2Y_3 + AlY_2
17#	1263	1365	Liquid/Liquid + Al_3Y_5
	1242	1146	Liquid + Al_3Y_5/Liquid + Al_3Y_5 + hcp_A3
	1180	1093	Liquid + Al_3Y_5 + hcp_A3/Al_3Y_5 + hcp_A3
	–	1058	Al_3Y_5 + hcp_A3/Al_3Y_5 + hcp_A3 + AlY_2
	–	1057	Al_3Y_5 + hcp_A3 + AlY_2/hcp_A3 + AlY_2
18#	1294	1439	Liquid/Liquid + Al_3Y_5
	1191	1237	Liquid + Al_3Y_5/Liquid + Al_3Y_5 + hcp_A3
	1162	1235	Liquid + Al_3Y_5 + hcp_A3/Al_3Y_5 + hcp_A3
	1154	1207	Al_3Y_5 + hcp_A3/Al_3Y_5 + hcp_A3 + AlY_2
	–	1160	Al_3Y_5 + hcp_A3 + AlY_2/AlY_2 + Al_3Y_5
	–	805	AlY_2 + Al_3Y_5/AlY_2 + Al_3Y_5 + Al_2Y_3
	–	798	AlY_2 + Al_3Y_5 + Al_2Y_3/AlY_2 + Al_2Y_3
19#	1269	1390	Liquid/Liquid + Al_3Y_5
	1225	1241	Liquid + Al_3Y_5/Liquid + Al_3Y_5 + hcp_A3
	–	1240	Liquid + Al_3Y_5 + hcp_A3/Al_3Y_5 + hcp_A3
	1170	1182	Al_3Y_5 + hcp_A3/Al_3Y_5 + AlY_2 + hcp_A3
	–	1141	Al_3Y_5 + AlY_2 + hcp_A3/AlY_2 + hcp_A3

(*continued on next page*)

Table 5 (*continued*)

Alloy No.	Measured temperature $T_{mea.}$ (K)	Calculated temperature $T_{cal.}$ (K)	Invariant reaction or phase boundary
20#	1197	1255	Liquid/Liquid + Al_3Y_5
	1183	1242	Liquid + Al_3Y_5/Liquid + Al_3Y_5 + hcp_A3
	–	1242	Liquid + Al_3Y_5 + hcp_A3/Al_3Y_5 + hcp_A3
	1150	1158	Al_3Y_5 + hcp_A3/Al_3Y_5 + AlY_2 + hcp_A3
	–	1127	Al_3Y_5 + AlY_2 + hcp_A3/AlY_2 + hcp_A3

[a] Data were measured under argon flow at a rate of 25 ml/min. Standard uncertainty u is $u(T_{mea.}) = 1$ K.
* Data are determined from the DSC cooling curves.

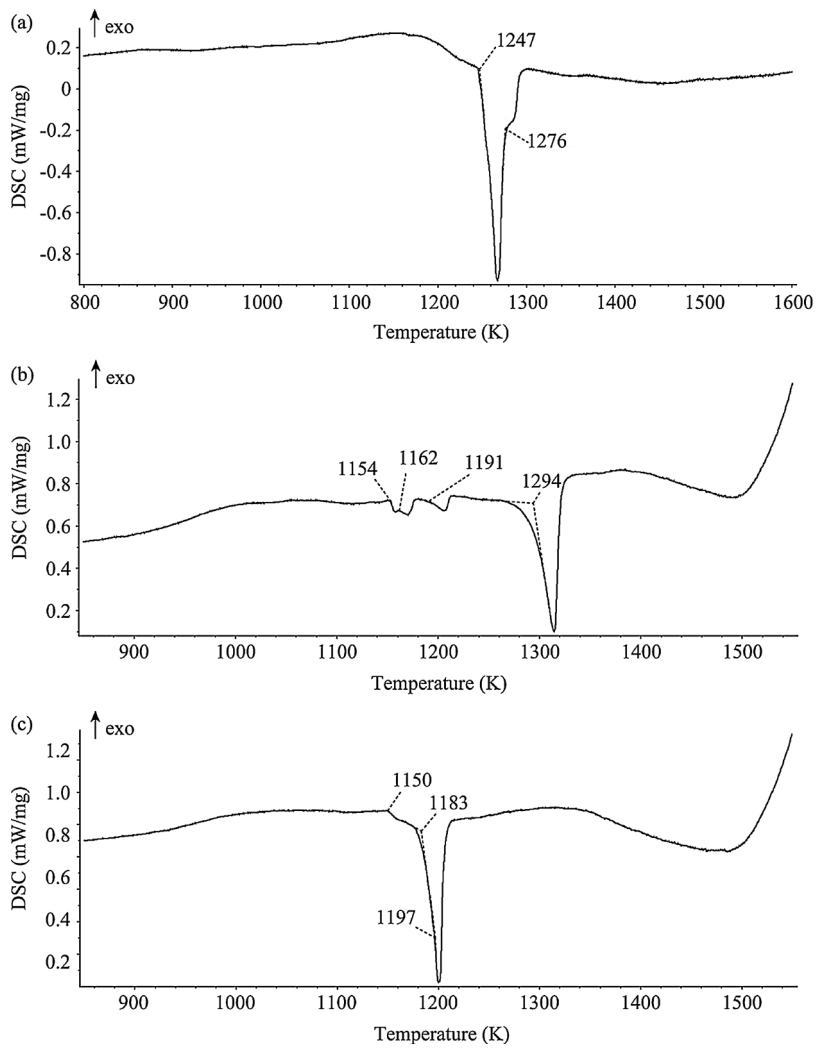

Fig. 3. DSC heating curves of the annealed samples: (a) 14#; (b) 18#; (c) 20#.

5. Results and discussion

5.1. Experimentation aspect

The equilibrium phase constitution and compositions of constituent phases determined by the SEM/EDX are summarized in Table 4. The ternary compound with a certain atomic ratio in the Al-rich corner, $Al_{75}Sc_{17}Y_8$, was discovered for the first time, the composition of which was within the homogeneity range of the Al_6ScY reported by Zalutskaya et al. [44]. The maximum solubility of Sc in the αAl_3Y phase was measured to be 5 at.%, while that of Y in the Al_3Sc phase 4 at.%. As has been mentioned earlier, there is an Al_3Y_5 metastable phase in the Al-Y binary system. After the addition of Sc, the element Sc could substitute the element Y of the Al_3Y_5 phase, making it stable over a large composition range in the ternary system. Since none of the alloy samples was positioned in the three-phase fields at the ends of the homogeneity range of the Al_3Y_5 phase, the exact maximum solubility could not be known. Yet, it exists between 37.5-56 at.% Y at least based on the tie-lines of the samples 14–16# and 18#. According to this experimental work, the Al-Y binary compounds all have certain solubilities for Sc and it is the same case for Y in the Al-Sc compounds excluding the $AlSc_2$ phase where the solubility of Y is unknown. The maximum solubilities of Sc in the Al_2Y_3, Al_3Y_5 and AlY_2 compounds are not determined by this work. Since there is no Al-Sc binary compound having the same crystal structure as them, the actual maximum solubility is supposed not to deviate a lot from the maximum value among the measured data. Nevertheless, further experimental investigation is still needed to obtain the exact value.

Fig. 1 presents the back scattered electron images of the annealed samples, namely 2#, 3#, 6#, 8#, 10#, 11#, 14# and 18#. The alloys

Table 6
Thermodynamic parameters of the Al-Sc-Y ternary system.

Phase	Thermodynamic parameters	Ref.
Liquid: (Al, Sc, Y)	${}^{0}L^{Liq}_{Al,Sc} = -131778 + 3.639T$	[56]
	${}^{1}L^{Liq}_{Al,Sc} = -20126 + 8.877T$	[56]
	${}^{2}L^{Liq}_{Al,Sc} = 50985 - 24.917T$	[56]
	${}^{0}L^{Liq}_{Al,Y} = -170000 + 37.5T$	this work
	${}^{1}L^{Liq}_{Al,Y} = -42100 + 14T$	this work
	${}^{2}L^{Liq}_{Al,Y} = 48000 - 12T$	this work
	${}^{0}L^{Liq}_{Sc,Y} = 13702 - 1.86T$	[43]
	${}^{1}L^{Liq}_{Sc,Y} = -0.5T$	[43]
	${}^{2}L^{Liq}_{Sc,Y} = 5000 - 0.341T$	[43]
fcc_A1: (Al, Sc, Y)$_1$(Va*)$_1$	${}^{0}L^{fcc.A1}_{Al,Sc:Va} = -109500 + 2.5T$	[56]
bcc_A2: (Al, Sc, Y)$_1$(Va*)$_3$	${}^{0}L^{bcc.A2}_{Al,Sc:Va} = -113589$	[56]
	${}^{0}L^{bcc.A2}_{Sc,Y:Va} = -1032 + 9.784T$	[43]
	${}^{2}L^{bcc.A2}_{Sc,Y:Va} = 1598 + 0.508T$	[43]
hcp_A3: (Al, Sc, Y)$_1$(Va*)$_{0.5}$	${}^{0}L^{hcp.A3}_{Al,Sc:Va} = -106397$	[56]
	${}^{0}L^{hcp.A3}_{Sc,Y:Va} = 1800+9.523T$	[43]
	${}^{2}L^{hcp.A3}_{Sc,Y:Va} = -488 + 1.112T$	[43]
Al_3Sc: (Al)$_3$(Sc, Y)$_1$	$G^{Al_3Sc}_{Al:Sc} = -156700+7.2T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	[56]
	$G^{Al_3Sc}_{Al:Y} = -166368 + 40T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	${}^{0}L^{Al_3Sc}_{Al:Sc,Y} = -33000$	this work
Al_2Sc: (Al)$_2$(Sc, Y)$_1$	$G^{Al_2Sc}_{Al:Sc} = -142500+8.8T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	[56]
	$G^{Al_2Sc}_{Al:Y} = -155732 + 26.9T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	${}^{0}L^{Al_2Sc}_{Al:Sc,Y} = -22000$	this work
AlSc: (Al)$_1$(Sc, Y)$_1$	$G^{AlSc}_{Al:Sc} = -94100+6.4T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	[56]
	$G^{AlSc}_{Al:Y} = -82408 + 12T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	${}^{0}L^{AlSc}_{Al:Sc,Y} = -28000$	this work
$AlSc_2$: (Al)$_1$(Sc)$_2$	$G^{AlSc_2}_{Al:Sc} = -112484 + 1.6T + {}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Sc}$	[56]
αAl_3Y: (Al)$_3$(Y, Sc)$_1$	$G^{\alpha Al_3Y}_{Al:Y} = -183600 + 42.7T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	$G^{\alpha Al_3Y}_{Al:Sc} = -140700 + 12T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	this work
	${}^{0}L^{\alpha Al_3Y}_{Al:Y,Sc} = -30000$	this work
βAl_3Y: (Al)$_3$(Y, Sc)$_1$	$G^{\beta Al_3Y}_{Al:Y} = -170000 + 27.9T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	$G^{\beta Al_3Y}_{Al:Sc} = -140700 + 16T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	this work
	${}^{0}L^{\beta Al_3Y}_{Al:Y,Sc} = -30000$	this work
Al_2Y: (Al, Y, Sc)$_2$(Al, Y, Sc)$_1$	$G^{Al_2Y}_{Al:Al} = 15000 + 3{}^{0}G^{fcc}_{Al}$	this work
	$G^{Al_2Y}_{Sc:Sc} = 15000 + 3{}^{0}G^{hcp}_{Sc}$	this work
	$G^{Al_2Y}_{Y:Y} = 15000 + 3{}^{0}G^{hcp}_{Y}$	this work
	$G^{Al_2Y}_{Al:Y} = -156300 + 21.9T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	$G^{Al_2Y}_{Y:Al} = 186300 - 21.9T + {}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Y}$	this work
	$G^{Al_2Y}_{Al:Sc} = -141742 + 19T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	this work
	$G^{Al_2Y}_{Sc:Al} = 171742 - 19T + {}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Sc}$	this work
	${}^{0}L^{Al_2Y}_{Al:Al,Y} = 32400$	this work
	${}^{0}L^{Al_2Y}_{Al,Y:Y} = 51100$	this work
	${}^{0}L^{Al_2Y}_{Al:Y,Sc} = -51000 + 5T$	this work
AlY: (Al)$_1$(Y, Sc)$_1$	$G^{AlY}_{Al:Y} = -89400+8.8T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Y}$	this work
	$G^{AlY}_{Al:Sc} = -80469 + 11T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	this work
	${}^{0}L^{AlY}_{Al:Y,Sc} = -52000$	this work
Al_2Y_3: (Al)$_2$(Y, Sc)$_3$	$G^{Al_2Y_3}_{Al:Y} = -200000 + 19.1T + 2{}^{0}G^{fcc}_{Al} + 3{}^{0}G^{hcp}_{Y}$	this work
	$G^{Al_2Y_3}_{Al:Sc} = -174233 + 16.5T + 2{}^{0}G^{fcc}_{Al} + 3{}^{0}G^{hcp}_{Sc}$	this work
	${}^{0}L^{Al_2Y_3}_{Al:Y,Sc} = -75500$	this work
Al_3Y_5: (Al)$_3$(Y, Sc)$_5$	$G^{Al_3Y_5}_{Al:Y} = -293323+23T + 3{}^{0}G^{fcc}_{Al} + 5{}^{0}G^{hcp}_{Y}$	this work
	$G^{Al_3Y_5}_{Al:Sc} = -247175 + 15T + 3{}^{0}G^{fcc}_{Al} + 5{}^{0}G^{hcp}_{Sc}$	this work
	${}^{0}L^{Al_3Y_5}_{Al:Y,Sc} = -156900$	this work
	${}^{1}L^{Al_3Y_5}_{Al:Y,Sc} = -15000$	this work

(*continued on next page*)

Table 6 (*continued*)

Phase	Thermodynamic parameters	Ref.
AlY_2: $(Al)_1(Y, Sc)_2$	$G^{AlY_2}_{Al:Y} = -109000+15.5T + {}^0G^{fcc}_{Al} + 2{}^0G^{hcp}_{Y}$	this work
	$G^{AlY_2}_{Al:Sc} = -101828 + 17T + {}^0G^{fcc}_{Al} + 2{}^0G^{hcp}_{Sc}$	this work
	${}^0L^{AlY_2}_{Al:Y,Sc} = -28500$	this work
	${}^1L^{AlY_2}_{Al:Y,Sc} = -10000$	this work
τ_1: $(Al)_{75}(Sc)_{17}(Y)_8$	$G^{\tau 1}_{Al:Sc:Y} = -4200000+300T + 75{}^0G^{fcc}_{Al} + 17{}^0G^{hcp}_{Sc} + 8{}^0G^{hcp}_{Y}$	this work

* Va denotes vacancy.

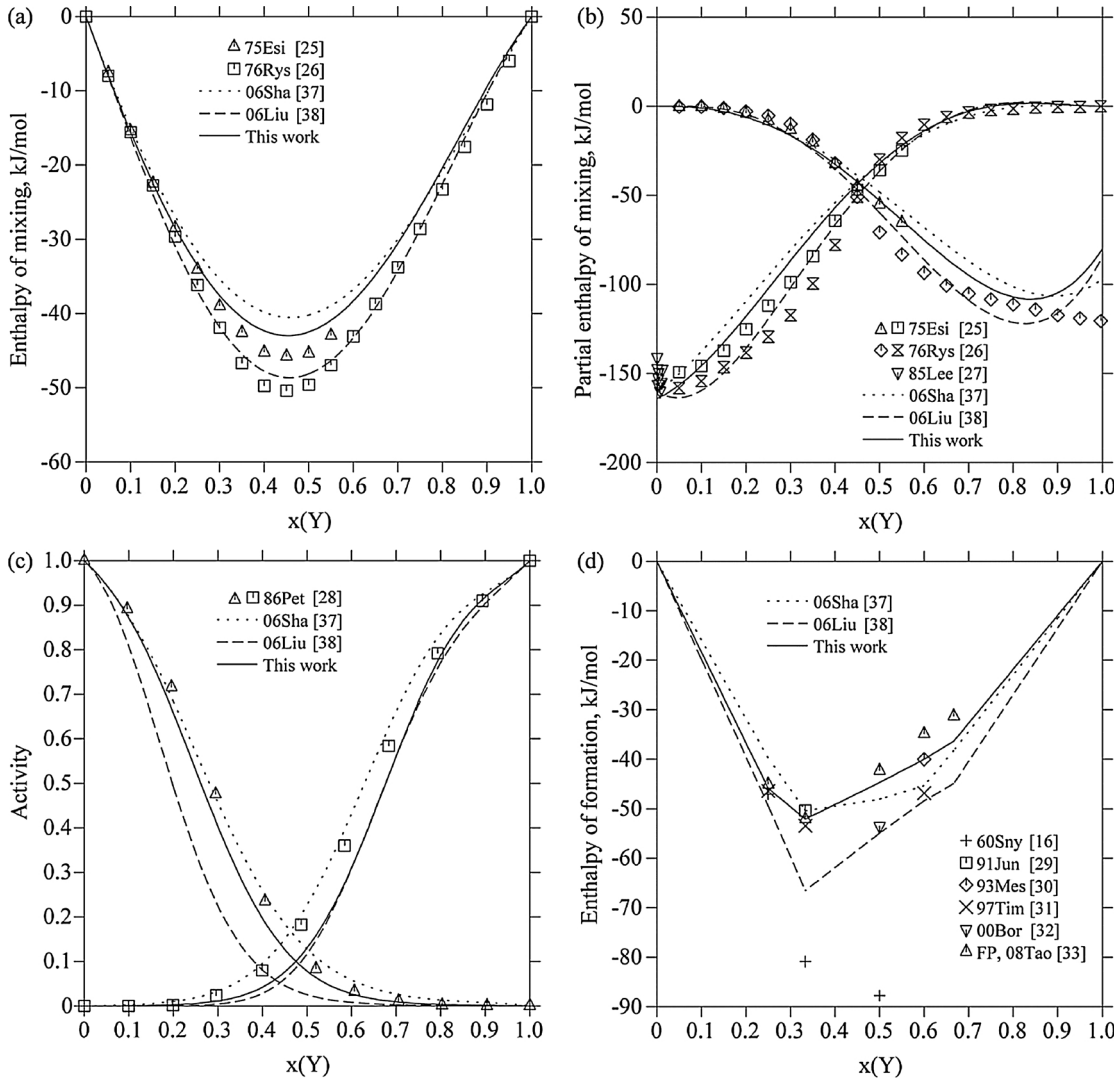

Fig. 4. Calculation results of the Al-Y binary system: (a) enthalpy of mixing of the liquid phase at 1873 K; (b) partial enthalpy of mixing of Al and Y at 1873 K; (c) activities of Al and Y at 1873 K; (d) enthalpy of formation of the compounds at 298 K.

2#, 6# and 8# are all constituted by three equilibrium phases, with one common constituent phase being $Al_{75}Sc_{17}Y_8$. The other alloys all consist of two equilibrium phases, which are labeled in the figure. It should be noted that there are also black phase in an irregular shape and white phase like flower petals in the micrographs of samples 10#, 11# and 14#, which are scandium oxide and yttrium oxide, respectively, according to the EDX results. The composition of the tiny black dots in the sample 18# could not be obtained accurately by the EDX analysis due to the small size. Yet, it is inferred to be scandium oxide based on the EDX result of the irregularly shaped black phase as mentioned above. The equilibrium phase constitution obtained by XRD is consistent with those by the SEM/EDX result. Fig. 2 shows the XRD patterns of some of the annealed alloys, namely 11#, 16#, 18# and 20#. Table 5 lists the phase transformation temperatures measured by DSC. They were all determined from the onset points of the endothermic peaks of the heating curves except those marked with an asterisk that were determined from the end points of the exothermic peaks of the cooling curves. Fig. 3 presents the DSC heating curves of the annealed samples 14#, 18# and 20#.

5.2. Modeling aspect

The optimization of the Al-Y binary system and the Al-Sc-Y ternary system was carried out on the Parrot module of the Thermo_calc

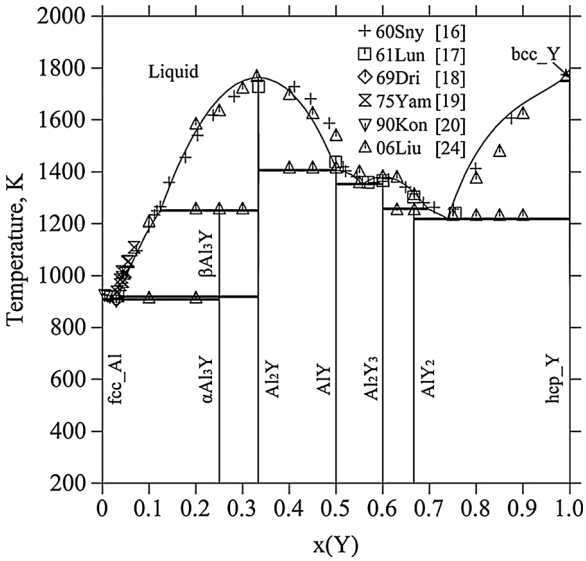

Fig. 5. Calculated phase diagram of the Al-Y binary system.

software developed by Sundman et al. [55]. The thermodynamic parameters of the Al-Sc and Sc-Y systems are adopted from our previous work [56] and the work of Wang et al. [43], respectively. Table 6 lists all the thermodynamic parameters of this ternary system.

Fig. 4 presents the optimized results of the Al-Y binary system. The calculated enthalpy of mixing of the liquid phase at 1873 K is shown in Fig. 4(a), along with previously optimized results [37,38]. From this figure, it can be seen that our calculation results just lie between those of Shakhshir and Medraj [37] and Liu et al. [38], and are closer to the experimental data of Esin et al. [25] than those of Ryss et al. [26]. The calculated partial mixing enthalpies of Al and Y at 1873 K are in a generally good agreement with the experimental data [25,26], as is shown in Fig. 4(b). It could be noticed that there is a big difference in the partial mixing enthalpy of Al in the melt with 95–100 at.% Y between any set of the mentioned calculation results and the data measured by Ryss et al. [26]. This may be attributed to the large amount of Y with a nature of being oxidized easily at high temperatures. Fig. 4(c) demonstrates the calculated activities of Al and Y at 1873 K. Since the superimposed data are obtained by calculation [28] rather than experimental measurement, they were set a low weight in this optimization work. The activity data were fitted as well as possible only on the basis of a good reproduction of the experimental data of integral and partial mixing enthalpy of the liquid phase. It can be seen from Fig. 4(c) that the optimized activities have a little difference from the reported data [28], which is thus acceptable. Fig. 4(d) presents the calculated standard enthalpy of formation of the Al-Y binary compounds, compared with the experimental data, first-principles calculation results and the optimized results from Shakhshir and Medraj [37] and Liu et al. [38]. With the first-principles results, it is easy to identify the reliable data from separate sets of experimental data. The formation enthalpies of the Al_2Y and AlY compounds measured by Snyder [16] were not taken into consideration during the optimization since they are too negative compared with others [29–32]. The calculated phase diagram of the Al-Y binary system has a good agreement with most of the experimental data, as is shown in Fig. 5. The differences between the optimized temperatures and the reported DTA data [24] are within 5 K for all the invariant reactions, except the ones in the Y-rich end, namely L + Al_2Y_3 ↔ AlY_2 and L ↔ AlY_2 + hcp_Y, with a difference of 8 K and 9 K, respectively. Table 7 summarizes the temperatures of the invariant reactions and the compositions of the equilibrium phases by calculation.

Integrating the thermodynamic parameters of the three constituent binary systems Al-Sc, Al-Y and Sc-Y, the Al-Sc-Y ternary system was thermodynamically optimized. Fig. 6 shows the calculated isothermal section at 873 K, where the EDX analyzed compositions of all the samples and the measured tie-lines and tie-triangles are superimposed. It can be seen that a good fit exists between the calculated phase relations and the measured ones.

The phase transformation temperatures of the samples were calculated and a comparison can be made in Table 5 between the calculated and the measured data. The invariant reactions or phase boundaries obtained by calculation are also listed and are corresponded to the measured temperatures in a tentative way. As can be seen from Table 5, the reactions with solid reactants or involving both βAl_3Y and αAl_3Y were not detected by the DSC tests, which might be attributed to the weak thermal effects. The vertical section at 9 at.% Sc calculated in this work is shown in Fig. 7, where the DSC data of the samples 19# and 2# are given for comparison. Another vertical section at 16 at.% Y is

Table 7
Calculated invariant reactions of the Al-Y binary system.

Reaction	T (K)	x (Y) in liquid	Ref.
Liquid ↔ fcc_Al + αAl_3Y	910		[24]
	913	0.029	[16]
	923	0.033	[17]
	906	0.032	[37]
	910	0.030	[38]
	908	0.032	This work
βAl_3Y ↔ αAl_3Y	917		[16]
	917		[38]
	919		This work
Liquid + Al_2Y ↔ βAl_3Y	1253		[24]
	1253	0.117	[16]
	1628	0.230	[17]
	1257	0.112	[37]
	1253	0.126	[38]
	1251	0.124	This work
Liquid ↔ Al_2Y	1763	0.333	[24]
	1758	0.333	[16]
	1728	0.333	[17]
	1751	0.333	[37]
	1763	0.333	[38]
	1763	0.333	This work
Liquid + Al_2Y ↔ AlY	1411		[24]
	1403	0.520	[16]
	1438		[17]
	1403	0.521	[37]
	1410	0.511	[38]
	1406	0.501	This work
Liquid ↔ AlY + Al_2Y_3	1354		[24]
	1361	0.580	[16]
	1358	0.570	[17]
	1359	0.573	[37]
	1356	0.562	[38]
	1353	0.557	This work
Liquid ↔ Al_2Y_3	1378	0.600	[24]
	1373	0.600	[16]
	1367	0.600	[17]
	1365	0.600	[37]
	1377	0.600	[38]
	1382	0.600	This work
Liquid + Al_2Y_3 ↔ AlY_2	1250		[24]
	1258	0.750	[16]
	1308	0.700	[17]
	1257	0.727	[37]
	1251	0.708	[38]
	1258	0.697	This work
Liquid ↔ AlY_2 + hcp_Y	1228	~0.750	[24]
	1233	0.753	[16]
	1243		[17]
	1250	0.736	[37]
	1227	0.737	[38]
	1219	0.738	This work
bcc_Y ↔ Liquid + hcp_Y	1751	0.978	[37]
	1752	0.979	[38]
	1751	0.979	This work

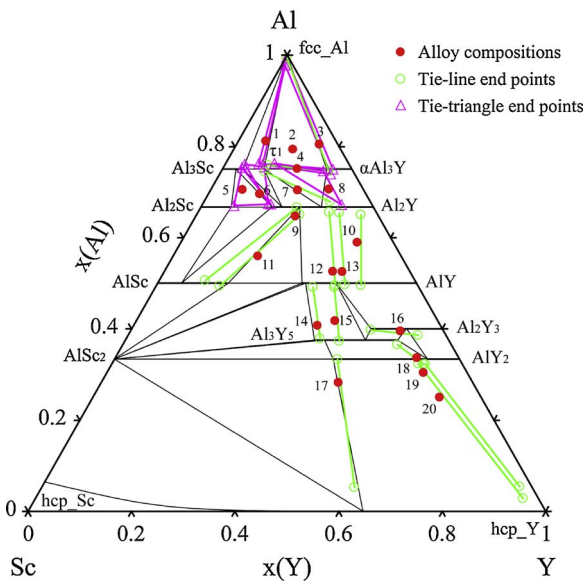

Fig. 6. Calculated isothermal section of the Al-Sc-Y system at 873 K along with the tie-lines and tie-triangles determined from individual samples in this work.

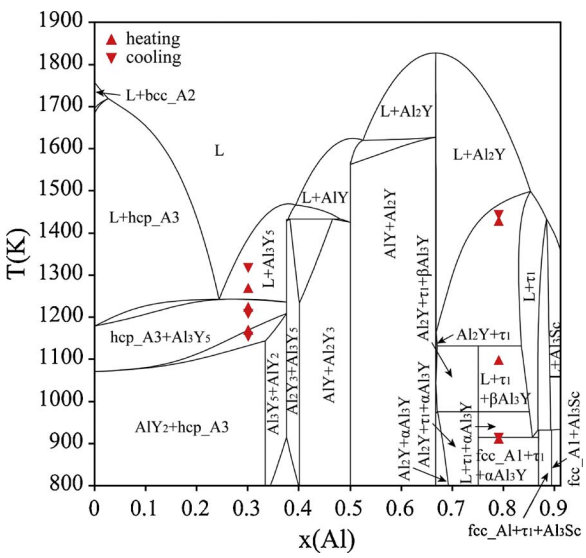

Fig. 7. Calculated vertical section of the Al-Sc-Y system at 9 at.% Sc.

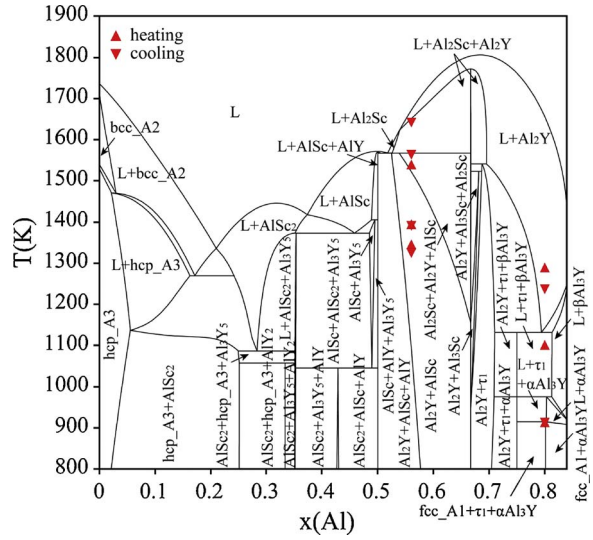

Fig. 8. Calculated vertical section of the Al-Sc-Y system at 16 at.% Y.

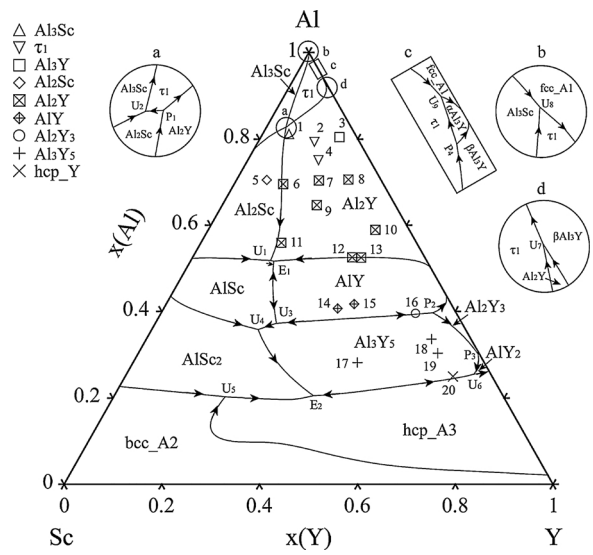

Fig. 9. Calculated liquidus projection of the Al-Sc-Y system together with the primary crystallization phases observed in the as-cast samples.

presented in Fig. 8, along with the DSC results of the samples 11# and 3#. From these two figures, one can notice that the calculated phase transition temperatures are basically consistent with the thermal analysis data.

The calculated liquidus projection of the Al-Sc-Y system is shown in Fig. 9. The primary crystallization phases obtained by calculation are marked in the figure and those observed from individual as-cast samples are given in the legend. It can be found that the samples 1# and 20# almost lie on the monovariant lines, which indicates that only a small difference exists between the calculated and experimentally observed primary crystallization phase. For the sample 16#, although it also lies on the monovariant line, the observed primary crystallization phase turns out to be Al_2Y_3, not the one on either side of the monovariant line, which shows slightly large difference between the calculated and experimental results. It is worth of being noted that the calculated primary crystallization phases for the samples 2#, 3# and 4# are all Al_2Y, while the observed ones are τ_1 phase for the samples 2# and 4#, and βAl_3Y phase for the sample 3#. This means that the calculated Al_2Y phase is over stable at high temperatures. As a matter of

Table 8
Calculated invariant reactions of the Al-Sc-Y ternary system.

Type	Reaction	T (K)
U_1	Liquid + $Al_2Sc \leftrightarrow Al_2Y$ + AlSc	1567
E_1	Liquid $\leftrightarrow Al_2Y$ + AlSc + AlY	1566
P_1	Liquid + Al_2Sc + $Al_2Y \leftrightarrow \tau_1$	1540
U_2	Liquid + $Al_2Sc \leftrightarrow \tau_1$ + Al_3Sc	1540
P_2	Liquid + AlY + $Al_3Y_5 \leftrightarrow Al_2Y_3$	1432
U_3	Liquid + AlY $\leftrightarrow$ AlSc + Al_3Y_5	1404
U_4	Liquid + AlSc $\leftrightarrow Al_3Y_5$ + $AlSc_2$	1372
U_5	Liquid + bcc_A2 $\leftrightarrow$ hcp_A3 + $AlSc_2$	1269
P_3	Liquid + Al_3Y_5 + $Al_2Y_3 \leftrightarrow AlY_2$	1261
U_6	Liquid + $Al_3Y_5 \leftrightarrow AlY_2$ + hcp_A3	1230
U_7	Liquid + $Al_2Y \leftrightarrow \tau_1$ + βAl_3Y	1131
E_2	Liquid $\leftrightarrow Al_3Y_5$ + hcp_A3 + $AlSc_2$	1086
P_4	Liquid + τ_1 + $\beta Al_3Y \leftrightarrow \alpha Al_3Y$	974
U_8	Liquid + $Al_3Sc \leftrightarrow \tau_1$ + fcc_A1	931
U_9	Liquid + $\tau_1 \leftrightarrow \alpha Al_3Y$ + fcc_A1	914

fact, the absolute value of the formation enthalpy of the Al_2Y phase re-optimized in this work is already smaller than all the optimized results reported in previous literatures and fits both the measured data and the first-principles calculation result very well. No matter how hard we tried to optimize the thermodynamic parameter of the Al_2Y phase, the primary crystallization surfaces of the τ_1 and βAl_3Y phases were not enlarged. The efforts made in modifying the thermodynamic parameters of the τ_1 and βAl_3Y phases are also in vain. Apart from the samples 2#, 3# and 4#, the calculated liquidus projection could account for the observed primary crystallization phases satisfactorily. Based on the calculation results, there are fifteen invariant reactions in total, as are given in Table 8.

6. Conclusions

The phase equilibria of the Al-Sc-Y system at 873 K have been determined using individual alloys. In the Al-rich corner, a ternary compound $Al_{75}Sc_{17}Y_8$ was discovered for the first time, the crystal structure of which is unknown. The Al_3Y_5 phase turns out to be stable in the ternary system, containing at least 37.5-56 at.% Y and a constant 37.5 at.% Al. All the Al-Sc and Al-Y binary compounds were found to have a certain solubility of the third component excluding the $AlSc_2$ phase, in which the solubility of Y was not determined in this experimental work. In addition, the exact maximum solubility of Sc in the compounds Al_2Y_3, Al_3Y_5 and AlY_2 need to be studied in the further experimental investigation. Based on the experimental information and the first-principles calculation results in this work, the Al-Sc-Y ternary system was optimized by the CALPHAD method, with the constituent Al-Y binary system re-optimized in advance. A reasonable thermodynamic description has finally been developed, since the calculated isothermal section at 873 K and the vertical sections at 9 at.% Sc and 16 at.% Y explain the experimental data well, and the calculated liquidus projection could account for most of the primary crystallization phases observed in the as-cast samples.

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References

[1] M.J. Li, Q.L. Pan, Y.J. Shi, X. Sun, H. Xiang, High strain rate superplasticity in an Al–Mg–Sc–Zr alloy processed via simple rolling, Mater. Sci. Eng. A 687 (2017) 298–305.
[2] X. Huang, Q.L. Pan, B. Li, Z.M. Liu, Z.Q. Huang, Z.M. Yin, Microstructure and mechanical properties of a 5754 aluminum alloy modified by Sc and Zr additions, J. Alloys Compd. 629 (2015) 197–207.
[3] J. Liu, P. Yao, N.Q. Zhao, C.S. Shi, H.J. Li, X. Li, D.S. Xi, S. Yang, Effect of minor Sc and Zr on recrystallization behavior and mechanical properties of novel Al–Zn–Mg–Cu alloys, J. Alloys Compd. 657 (2016) 717–725.
[4] A. Kumar, A.K. Mukhopadhyay, K.S. Prasad, Superplastic behaviour of Al–Zn–Mg–Cu–Zr alloy AA7010 containing Sc, Mater. Sci. Eng. A 527 (2010) 854–857.
[5] N.A. Belov, A.N. Alabin, I.A. Matveeva, Optimization of phase composition of Al–Cu–Mn–Zr–Sc alloys for rolled products without requirement for solution treatment and quenching, J. Alloys Compd. 583 (2014) 206–213.
[6] C.C. Shi, L. Zhang, G.H. Wu, X.L. Zhang, A.T. Chen, J.S. Tao, Effects of Sc addition on the microstructure and mechanical properties of cast Al-3Li-1.5Cu-0.15Zr alloy, Mater. Sci. Eng. A 680 (2017) 232–238.
[7] P.W. Voorhees, Scandium overtakes zirconium, Nat. Mater. 5 (2006) 435–436.
[8] K.E. Knipling, D.C. Dunand, D.N. Seidman, Criteria for developing castable, creep-resistant aluminum-based alloys—a review, Z. Metallkd. 97 (2006) 246–265.
[9] W. Kang, H.Y. Li, S.X. Zhao, Y. Han, C.L. Yang, G. Ma, Effects of homogenization treatments on the microstructure evolution, microhardness and electrical conductivity of dilute Al–Sc–Zr–Er alloys, J. Alloys Compd. 704 (2017) 683–692.
[10] A.V. Pozdniakov, V. Yarasu, R.Y. Barkov, O.A. Yakovtseva, S.V. Makhov, V.I. Napalkov, Microstructure and mechanical properties of novel Al–Mg–Mn–Zr–Sc–Er alloy, Mater. Lett. 202 (2017) 116–119.
[11] A.D. Luca, D.C. Dunand, D.N. Seidman, Mechanical properties and optimization of the aging of a dilute Al–Sc–Er–Zr–Si alloy with a high Zr/Sc ratio, Acta Mater. 119 (2016) 35–42.
[12] C. Monachon, M.E. Krug, D.N. Seidman, D.C. Dunand, Chemistry and structure of core/double-shell nanoscale precipitates in Al–6.5Li–0.07Sc–0.02Yb (at.%), Acta Mater. 59 (2011) 3398–3409.
[13] M.E. Krug, A. Werber, D.C. Dunand, D.N. Seidman, Core-shell nanoscale precipitates in Al–0.06 at.% Sc microalloyed with Tb, Ho, Tm or Lu, Acta Mater. 58 (2010) 134–145.
[14] M.E. van Dalen, D.C. Dunand, D.N. Seidman, Nanoscale precipitation and mechanical properties of Al–0. 06% Sc alloys microalloyed with Yb or Gd, J. Mater. Sci. 41 (2006) 7814–7823.
[15] R.A. Karnesky, M.E. van Dalen, D.C. Dunand, D.N. Seidman, Effects of substituting rare-earth elements for scandium in a precipitation-strengthened Al–0. 08 at.% Sc alloy, Scripta Mater. 55 (2006) 437–440.
[16] R.L. Snyder, PhD Thesis, Yttrium–Aluminum Alloy Studies, Iowa State University, Ames, USA, 1960.
[17] C.E. Lundin, D.T. Klodt, Phase equilibria in the yttrium–aluminum system, Trans. ASM 54 (1961) 168–175.
[18] M.E. Drits, E.S. Kadaner, N.D. Shoa, Solid solubility of rare earth metals in aluminum, Russ. Metall. 1 (1969) 113–117.
[19] L.F. Yamshchikov, V.I. Kober, V.A. Lebedev, I.F. Nichkov, S.P. Raspopin, Thermodynamic properties of aluminum-rich yttrium-aluminum alloys, Russ. J. Phys. Chem. 49 (1975) 1730–1732.
[20] V.I. Kononenko, S.V. Golubev, Phase diagrams of the binary systems of aluminum with lanthanum cerium, praseodymium, neodymium, samarium, europium, ytterbium, scandium, and yttrium, Izv. Akad. Nauk SSSR Metally 2 (1990) 197–199.
[21] R. Richter, Z. Altounian, J.O. Ström-Olsen, Y_5Al_3: a new Y-Al compound, J. Mater. Sci. 22 (1987) 2983–2986.
[22] L.M. Zeng, S.Y. Wang, The 800 K isothermal section of the Y–Al–Sb phase diagram, J. Alloys Compd. 351 (2003) 176–179.
[23] J. She, Y.Z. Zhan, Z.H. Hu, C.L. Li, J. Hu, Y. Du, H.H. Xu, Experimental study of Al–Zr–Y system phase equilibria at 773 K, J. Alloys Compd. 497 (2010) 118–120.
[24] S.H. Liu, Y. Du, H.H. Xu, C.Y. He, J.C. Schuster, Experimental investigation of the Al–Y phase diagram, J. Alloys Compd. 414 (2006) 60–65.
[25] Y.O. Esin, P.V. Gel'd, M.S. Petrusheveskii, N.P. Bobrov, G.H. Ryss, V.V. Sokolov, I.P. Pazdnikov, A.Ya. Dubrovskii, Enthalpies of formation of molten binary aluminum alloys with yttrium, zirconium, vanadium and chromium, Splavy Redk. Met. Osobymi Fiz-Khim. Svoistvami (1975) 177–180.
[26] G.M. Ryss, Y.O. Esin, A.I. Stroganov, P.V. Gel'd, Enthalpies of formation of yttrium-aluminum molten alloys, Zh. Fiz. Khim 50 (1976) 985–986.
[27] J.-J. Lee, F. Sommer, The determination of the partial enthalpies of mixing of aluminum-rich alloy melts by solution calorimetry, Z. Metallkd. 76 (1985) 750–754.
[28] M.S. Petrusheveskii, G.M. Ryss, Calculations of the activities of components in liquid binary alloys of yttrium with aluminum and silicon, Zh. Fiz. Khim. 60 (1986) 1532–1535.
[29] W.-G. Jung, O.J. Kleppa, L. Topor, Standard molar enthalpies of formation of PdAl, PtAl $ScAl_{1.78}$, YAl_2 and $LaAl_2$, J. Alloys Compd. 176 (1991) 309–318.
[30] S.V. Meschel, O.J. Kleppa, Standard enthalpies of formation of 4d aluminides by direct synthesis calorimetry, J. Alloys Compd. 191 (1993) 111–116.
[31] V.S. Timofeev, A.A. Turchanin, A.A. Zubkov, I.A. Tomilin, Enthalpies of formation for the Al–Y and Al–Y–Ni intermetallic compounds, Thermochim. Acta 299 (1997) 37–41.
[32] G. Borzone, A. Ciccioli, P.L. Cignini, M. Ferrini, D. Gozzi, Thermodynamics of the YAl–YAl_2 system, Intermetallics 8 (2000) 203–212.
[33] X.M. Tao, PhD Thesis, First-Principles Calculations of the Thermodynamic Properties of Rare Earths-aluminum and Rare Earths-Aluminum Alloys, Central South University, Changsha, China, 2008.
[34] Q.S. Ran, H.L. Lukas, G. Effenberg, G. Petzow, A thermodynamic optimization of the Al–Y system, J. Less-Common Met. 146 (1989) 213–222.
[35] J. Gröbner, H.-L. Lukas, F. Aldinger, Thermodynamic calculations in the Y–Al–C system, J. Alloys Compd. 220 (1995) 8–14.
[36] H.L. Lukas, System Al–Y, COST-507, in: I.A.T. Ansara, M.H.R. Dinsdale (Eds.), Thermochemical Database for Light Metal Alloys, European Communities, Luxembourg, 1998, pp. 99–102.
[37] S.A. Shakhshir, M. Medraj, Computational thermodynamic model for the Mg–Al–Y System, J. Phase Equilib. Diff. 27 (2006) 231–244.
[38] S.H. Liu, Y. Du, H.L. Chen, A thermodynamic reassessment of the Al–Y system, CALPHAD 30 (2006) 334–340.
[39] Y.-B. Kang, A.D. Pelton, P. Chartrand, C.D. Fuerst, Critical evaluation and thermodynamic optimization of the Al–Ce, Al–Y, Al–Sc and Mg–Sc binary systems, CALPHAD 32 (2008) 413–422.
[40] B.J. Beaudry, A.H. Daane, The scandium–yttrium and scandium–zirconium system, Trans. Metall. Soc. AIME 227 (1963) 865–868.
[41] K.A. Gschneidner Jr., F.W. Calderwood, The Sc–Y (scandium-yttrium) system, Bull. Alloy Phase Diagr. 4 (1983) 171–172.
[42] A.K. Niessen, F.R. de Boer, R. Boom, P.F. de Châtel, W.C.M. Mattens, A.R. Miedema, Model predictions for the enthalpy of formation of transition metal alloys II, CALPHAD 7 (1983) 51–70.
[43] S.L. Wang, C.P. Wang, X.J. Liu, K. Ishida, Thermodynamic assessments of the Ag–Y and Sc–Y systems, J. Alloys Compd. 476 (2009) 187–192.
[44] O.I. Zalutskaya, V.R. Ryabov, I.I. Zalutsky, The x-ray analysis of aluminium-rich ternary alloys of aluminium and rare-earth metals of the yttrium subgroup and

scandium, Dopov. Akad. Nauk Ukr. RSR, Ser. A 3 (1969) 255–259.
[45] P. Villars, P. Villars, A. Prince, H. Okamoto (Eds.), Handbook of Ternary Alloy Phase Diagrams (Vol. 1), ASM International, Materials Park, OH, USA, 1997.
[46] P. Villars, A. Prince, H. Okamoto (Eds.), Handbook of Ternary Alloy Phase Diagrams (Vol. 2), ASM International, Materials Park, OH, USA, 1997.
[47] A.T. Dinsdale, SGTE Unary Database, Version 4.4, (2001).
[48] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B 54 (1996) 11169–11186.
[49] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semi-conductors using a plane-wave basis set, Comput. Mater. Sci. 6 (1996) 15–50.
[50] P.E. Blöchl, Projector augmented-wave method, Phys. Rev. B 50 (1994) 17953–17979.
[51] G. Kresse, D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B 59 (1999) 1758–1775.
[52] J.P. Perdew, J.A. Chevary, S.H. Vosko, K.A. Jackson, M.R. Pederson, D.J. Singh, C. Fiolhais, Atoms, molecules, solids and surfaces: applications of the generalized gradient approximation for exchange and correlation, Phys. Rev. B 46 (1992) 6671–6687.
[53] J.P. Perdew, Y. Wang, Accurate and simple analytic representation of the electron-gas correlation energy, Phys. Rev. B 45 (1992) 13244–13249.
[54] H. Bo, L.B. Liu, J.L. Hu, Z.P. Jin, Experimental study and thermodynamic assessment of the Zr–Al–Gd system, Thermochim. Acta 609 (2015) 36–48.
[55] B. Sundman, B. Jansson, J.O. Andersson, The thermo-calc databank system, CALPHAD 9 (1985) 153–190.
[56] H. Bo, L.B. Liu, Z.P. Jin, Thermodynamic analysis of Al–Sc, Cu–Sc and Al–Cu–Sc systems, J. Alloys Compd. 490 (2010) 318–325.